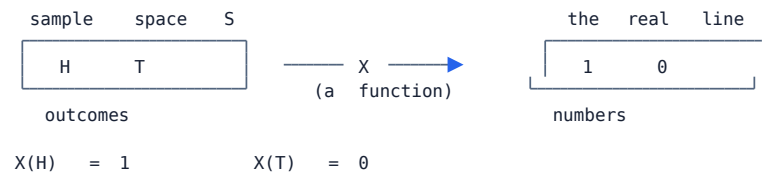


02 RANDOM VARIABLES (notes p051-p062)

1. What a random variable IS (p051)

A random variable is a FUNCTION whose domain is the sample space and whose range is the set of real numbers.
It is NOT the outcome. It is the NUMBER you attach to it.



2. Example 1: Bernoulli (p052)

toss one fair coin, $S = \{H, T\}$
define $X(H) = 1$, $X(T) = 0$
DEFINITION: any rv whose only values are 0 and 1 is a
BERNOULLI random variable.

Bernoulli is the atom of the binomial: one trial, success = 1, failure = 0.

3. Example 2: two coins (p053)

$S = \{HH, HT, TH, TT\}$
 $X = \text{number of heads}$

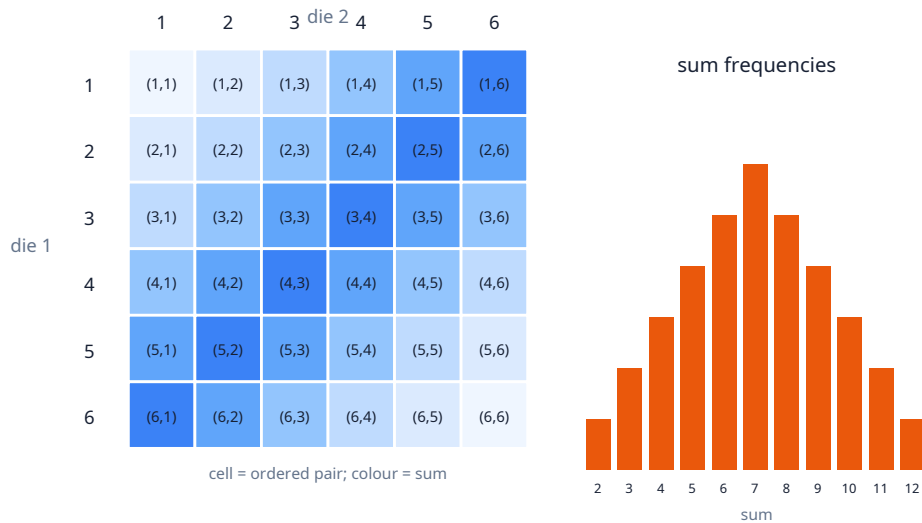
outcome	X value
HH	2
HT	1
TH	1
TT	0

$X(HH)=2,$ $X(TT)=0,$ $X(TH)=1,$ $X(HT)=1$

Notice HT and TH give the SAME X. Different outcomes, same value. That collapse is how a pmf is born (next file).

4. Example 3: dice sum (p054-p055)

Two dice: 36 ordered outcomes



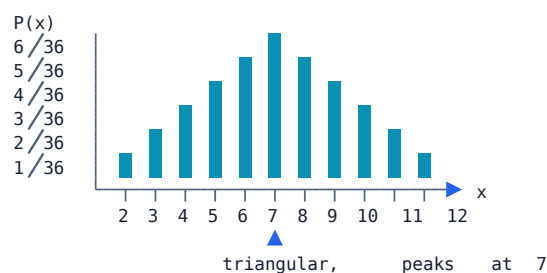
Ordered outcomes and the distribution of their sum

X = sum of two fair dice

x	outcomes producing it	count	P(X=x)
2	(1,1)	1	1/36
3	(1,2) (2,1)	2	2/36
4	(1,3) (2,2) (3,1)	3	3/36
5	(1,4) (2,3) (3,2) (4,1)	4	4/36
6	5 outcomes	5	5/36
7	6 outcomes	6	6/36
8	5 outcomes	5	5/36
9	4 outcomes	4	4/36
10	3 outcomes	3	3/36
11	(5,6) (6,5)	2	2/36
12	(6,6)	1	1/36
	Σ	36	36/36 = 1 ✓

<- peak

The pmf as a picture:



The identity the slide writes:

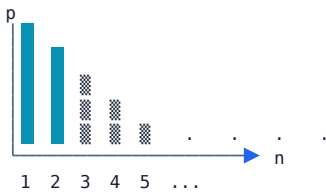
12		
1 = P(u	(X=n)) = Σ	P(X=n)
n=2	n=2	

5. Example 4: two coins as $Y = \text{heads}$ (p056)

$P(Y=0) = P(TT)$	$= 1/4$
$P(Y=1) = P(HT, TH)$	$= 2/4$
$P(Y=2) = P(HH)$	$= 1/4$
sum = 1	✓

6. Example 5: geometric (p057-p059) - counts flips until first head

N = number of flips needed to get the first head	
$P(N=1) = P(H)$	$= p$
$P(N=2) = P(TH)$	$= (1-p)p$
$P(N=3) = P(TTH)$	$= (1-p)^2 p$
...	
$P(N=n) = P(T...TH)$	$= (1-p)^{n-1} p$



<- each bar smaller by factor $(1-p)$

The sum verification (infinite geometric series, common ratio $(1-p)$):

$$\sum (1-p)^{n-1} p = p / (1 - (1-p)) = p/p = 1 \checkmark$$

This is the GEOMETRIC distribution. Mean = $1/p$.
It has NO teaching slide for its mean, but the exam asks it (assignment 1 short Q4: $E = 2$).

7. Example 6: car lifetime (p060) - the bridge to continuous

X = lifetime of a car, assumed to take any value in an interval (a,b)
 - not countable, cannot be listed as a sequence
 -> NOT a discrete rv

8. Discrete vs continuous, the definitions (p061-p063)

DISCRETE	CONTINUOUS
finite OR countably infinite	all values in an interval
can be arranged as a sequence	CANNOT be listed as a sequence
1st, 2nd, 3rd, ...	
	$P(X = \text{any single } c) = 0$

The formal continuous definition has TWO parts:

1. possible values = all numbers in one interval (or a union of intervals)
e.g. $[0,10] \cup [20,30]$
2. NO single value has positive probability: $P(X = c) = 0$

WHY $P(X=c)=0$ FOR CONTINUOUS

$$P(X=c) = \int \text{from } c \text{ to } c \text{ of } f(x) \, dx = 0$$

the width is zero, an area of zero width is zero

-> endpoints never matter: $P(a < X < b) = P(a \leq X \leq b)$

9. The summary table to memorise

	DISCRETE	CONTINUOUS
support	list of values	interval(s)
tool	Σ (sum)	$\int$ (integral)
$P(X=x)$	$p(x) > 0$	0 always
$E(X)$	$\Sigma x \cdot p(x)$	$\int x \cdot f(x) \, dx$
$\text{Var}(X)$	$\Sigma (x-\mu)^2 p(x)$	$\int (x-\mu)^2 f(x) \, dx$
named	binomial,	uniform, normal,
examples	Poisson, geo	exponential

Next file builds the pmf and cdf (the two functions that describe any rv).